

TRUSTIA AI

Autonomous. Deterministic. Mission-Critical.

NATO Innovation Fund (NIF) Seed Investment Pitch Deck

Emerging & Disruptive Technology: Autonomy

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1. THE PROBLEM

NATO Allied Forces face a critical autonomous ground vehicle capability gap:

- Tactical Convoy Vulnerability: Over 60% of NATO combat casualties in asymmetric theaters (Afghanistan, Iraq, Syria) were caused by IEDs targeting manned convoys and logistics vehicles.
- GPS-Denied Environments: Peer adversaries (electronic warfare, GPS jamming) render satellite-dependent navigation systems ineffective in contested zones.
- Prohibitive Costs: Current Western L4 autonomy platforms (Oshkosh TerraMax, Milrem THeMIS) cost \$500K–\$2M+ per vehicle with 18–24 month integration timelines.
- □ OEM Lock-In: Existing solutions are hardware-locked to specific vehicle platforms, preventing rapid cross-fleet deployment across NATO's diverse armored inventory.

2. THE SOLUTION

Trustia AI delivers a hardware-agnostic, 15-minute retrofit kit that converts ANY standard military or commercial ground vehicle into a fully SAE Level-4 autonomous platform.

Deterministic AI Core: 16,000 lines of original, non-stochastic autonomy software — no black-box neural network decision-making in safety-critical path planning.

15-Minute Retrofit: Plug-and-play CAN-Bus SAE J1939 integration. No cutting, welding, or mechanical modification to the host vehicle.

GPS-Denied Navigation: 400Hz Extended Kalman Filter (ESKF) state estimation with LiDAR SLAM — full autonomous capability under electronic warfare conditions.

IED/Mine Detection: Deep learning threat classification pipeline integrated into the autonomy loop for real-time route deviation.

Tactical C2 Console: Desktop Command & Control interface for multi-vehicle fleet orchestration, mission planning, and real-time telemetry.

3. TECHNICAL ARCHITECTURE

SUBSYSTEM	SPECIFICATION	NATO RELEVANCE
Perception	3D LiDAR Point Cloud + Stereo Camera Fusion	All-weather, day/night obstacle detection
Localization	400Hz ESKF + LiDAR SLAM (GPS-denied)	EW-resilient navigation in contested zones
Path Planning	Hybrid A* with Reeds-Shepp curves, <15ms	Real-time tactical route computation
Motion Control	Pure Pursuit + Stanley lateral controller	Smooth trajectory tracking on rough terrain
Vehicle Interface	CAN-Bus SAE J1939 (MIL-STD compatible)	Universal NATO vehicle fleet compatibility
Threat Detection	YOLOv8 + Custom IED/Mine classifier	Force protection & route clearance
C2 Integration	JAUS-compliant Tactical C2 Console	Interoperability with NATO C4ISR systems
Testing	1,281/1,281 Unit & Integration Tests (100%)	Deterministic verification & validation

4. COMPETITIVE ADVANTAGE

FEATURE	TRUSTIA AI	OSHKOSH TERRAMAX	MILREM THEMIS	ADASTEC
Architecture	Deterministic	Probabilistic	Probabilistic	Probabilistic
Vehicle Agnostic	☒ Any CAN-Bus	☐ JLTV Only	☐ THeMIS Only	☐ Bus Only
Retrofit Time	15 minutes	6–12 months	N/A (new build)	3–6 months
GPS-Denied Nav	☐ Full SLAM	☐ Partial	☐ Partial	☐ No
IED Detection	☐ Integrated	☐ No	☐ Add-on	☐ No
Unit Cost	\$15K–\$25K	\$500K–\$2M	\$200K–\$500K	\$100K+
Test Coverage	100% (1,281)	Classified	Classified	Undisclosed
NATO Member HQ	☐ Turkey	☐ USA	☐ Estonia	☐ Turkey

5. MARKET OPPORTUNITY

MARKET SEGMENT	TAM (2030)	CAGR
Global Military Autonomous Ground Vehicles	\$18.2 Billion	12.8%
NATO Defense Autonomous Systems Procurement	\$6.4 Billion	15.2%
Commercial Fleet Retrofit Autonomy	\$42.7 Billion	22.1%
IED/Mine Detection & Route Clearance	\$3.8 Billion	9.4%

6. BUSINESS MODEL

Hardware Retrofit Kit Sales: \$15,000–\$25,000 per vehicle (one-time). Includes compute unit, sensors, CAN-Bus harness, and mounting hardware.

Annual Software License (SaaS): \$3,000–\$5,000 per vehicle/year. Includes OTA updates, C2 console access, threat database sync, and 24/7 fleet monitoring.

Defense Consortium R&D;: Joint TÜBİTAK 1501 / Horizon Europe / NIF-funded collaborative R&D; programs with NATO prime contractors.

Training & Integration: \$50,000–\$150,000 per contract. On-site operator training, vehicle integration support, and mission planning workshops.

7. TRACTION & CREDENTIALS

CREDENTIAL	DETAIL
SSB Defense Industry Exam	100/100 Perfect Score (Verification: L2zPtN4X1ZJ)
KOSGEB Certification	Advanced Entrepreneur (KSB01UGE0115153370)
TÜBİTAK ARBİS	National Researcher Registry (TBTK-0229-6571)
ASELSAN	Registered Defense Supplier Portal
İTO BTM Incubation	2026 Term II Pre-Incubation (Istanbul Chamber of Commerce)
Codebase	16,000 lines original deterministic autonomy architecture
Test Coverage	1,281/1,281 automated tests — 100% pass rate
Global Applications	Seedcamp, a16z, Thiel Fellowship, Techstars, 500 Global submitted

8. FOUNDING TEAM

ROLE	NAME	BACKGROUND
Founder & CEO Systems Architect	Murat Furkan Bayram (Age 17)	Designed and built the entire 16,000-line L4 autonomy stack using next-generation
Hardware Engineer (Candidate)	Denizcan Özcan (İÜC EEE, 3.44 GPA)	ASELSAN embedded systems experience. TEKNOFEST Robotaxi Finalist. Special

9. THE ASK — NIF SEED INVESTMENT

Investment Sought: €1,000,000 – €3,000,000 Seed Round

USE OF FUNDS	ALLOCATION	TIMELINE
Hardware Prototype (LiDAR, Jetson, Sensors)	35%	Q4 2026 – Q1 2027
Field Testing on NATO-standard Vehicles	25%	Q1 – Q2 2027
Engineering Team Expansion (3–5 engineers)	20%	Q4 2026 – Q2 2027
Regulatory & NATO STANAG Certification	10%	Q2 – Q3 2027
Operations, Legal & IP Protection	10%	Ongoing

10. CONTACT

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Protecting Allied Forces Through Intelligent Autonomy

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